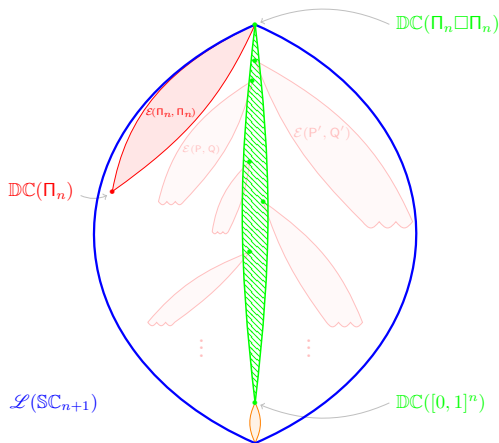


How many generalized permutahedra are there?

Germain Poullot with Georg Loho & Arnau Padrol



24 March 2026, Combinatorial CoWorkspace
arXiv:2510.03177

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Yes! Prepare your exponentials!

1 Generalized permutohedra, submodular cone

- 4 definitions
- Deformations

2 GP-sums and equality sets

- $\text{SC}_{n+1} \simeq \text{SC}_n^2$ more or less
- Drawing $\mathcal{L}(\text{SC}_n)^2$ inside $\mathcal{L}(\text{SC}_{n+1})$

3 Applications (now, we can count stuff)

- Rays
- f -vector, total number of faces, non-simplicial faces

Generalized permutohedra, submodular cone

Generalized permutohedra

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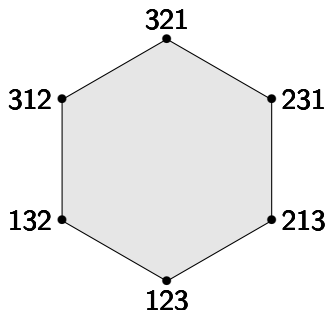
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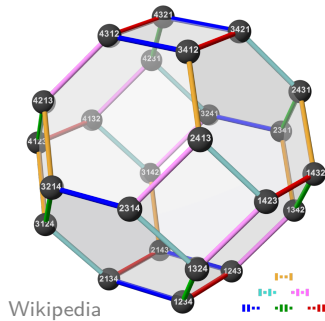
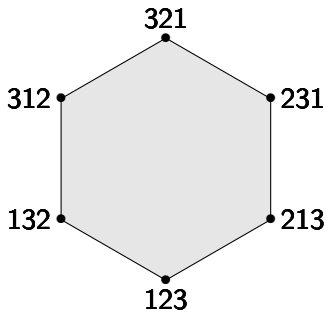
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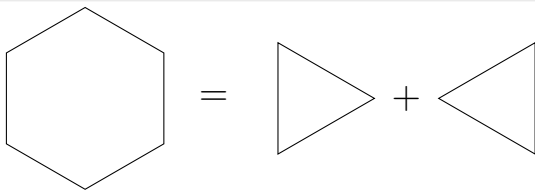
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(weak) Minkowski summands

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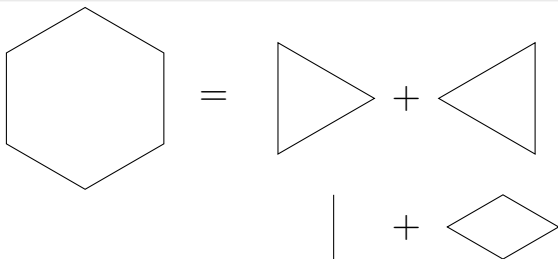
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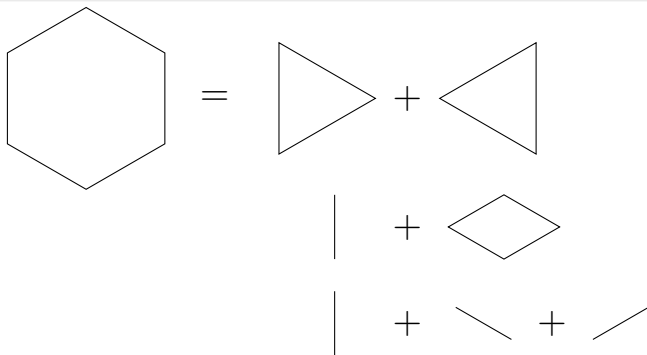
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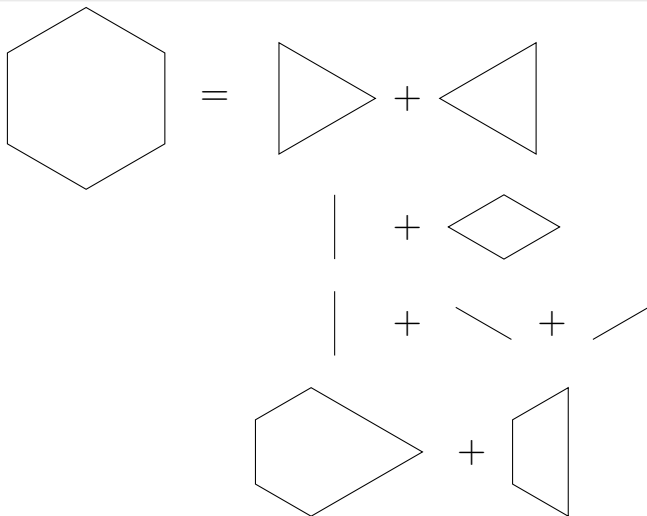
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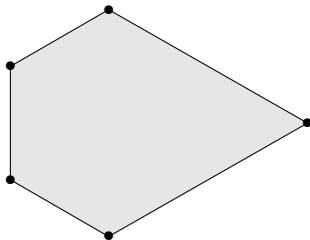
Fan coarsening

$\mathcal{N}_P$: Normal fan of polytope P

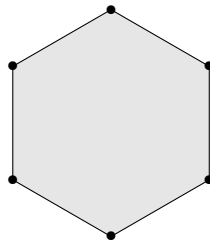
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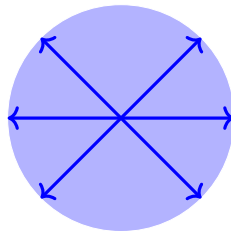
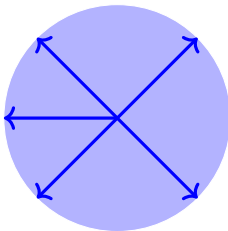
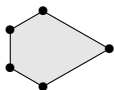
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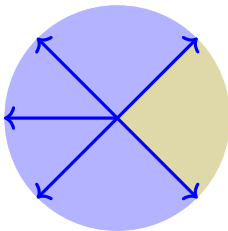
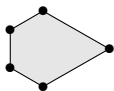
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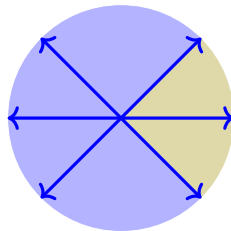
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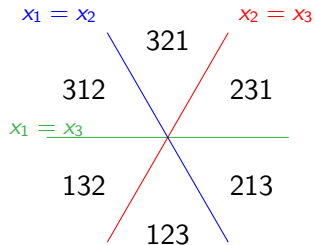
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Braid fan

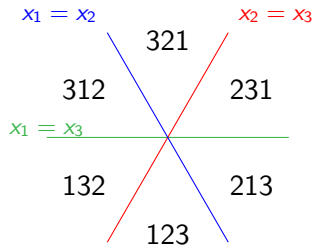
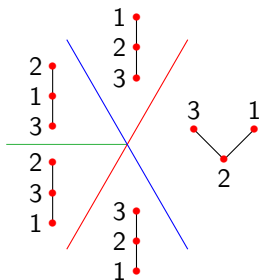
Definition

Braid fan: arrangement of hyperplanes $H_{i,j} := \{\mathbf{x} ; x_i = x_j\}$



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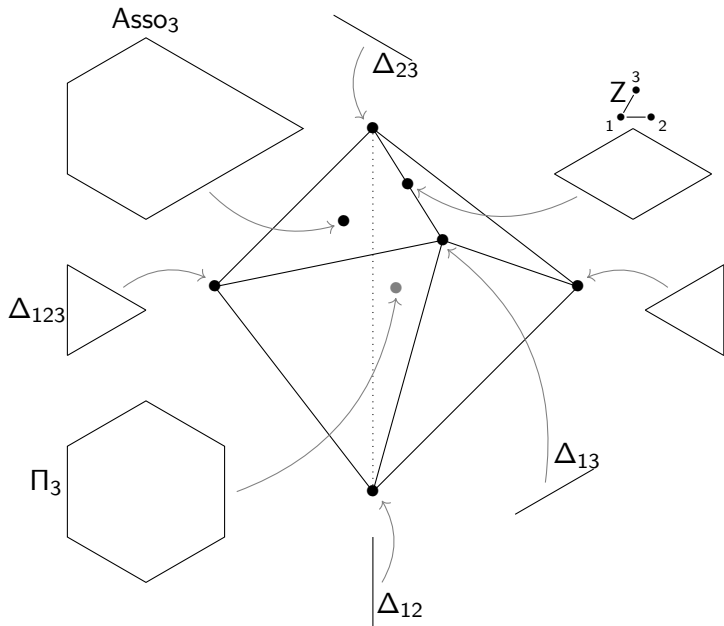
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Submodular cone $\text{SC}_n := \{\text{all generalized } n\text{-permutohedra}\}$



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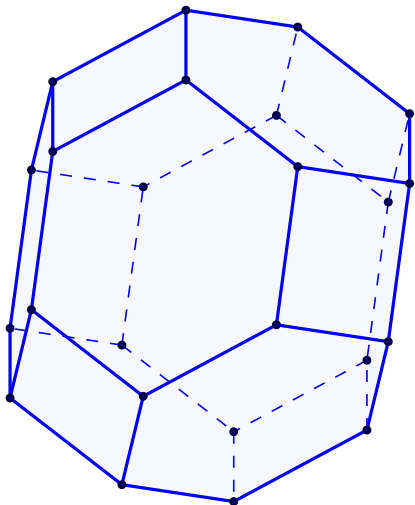
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P and P' are *equivalent*, denoted $P \sim P'$, iff $P' \trianglelefteq P$ and $P \trianglelefteq P'$

Deformations of Π_4



Permutahedron Π_4

Sequence of deformations of Π_4

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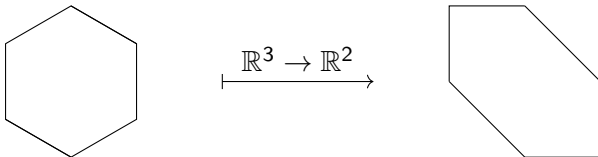
$$\dim \mathbb{SC}_n = 2^n - 1, \quad \#\text{facets } \mathbb{SC}_n = \binom{n}{2} 2^{n-2},$$

Edmonds '70: $\#\text{rays } \mathbb{SC}_n =$ **please compute it!**

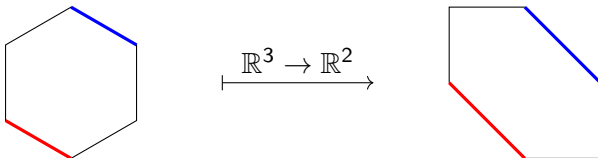
Some faces are “known”: associahedron, graphical zonotopes (arXiv:2111.12422), nestohedra (arXiv:2109.09200), etc.

GP-sums and equality sets

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 Now, edges directions are $\mathbf{e}_i$ and $\mathbf{e}_i - \mathbf{e}_j$ for $1 \leq i < j \leq n$

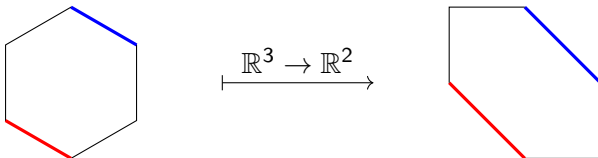


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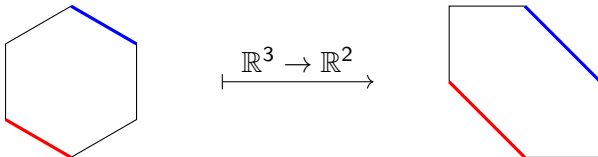
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*Deformed permutahedra are **uniquely** determined by their pair of top and bottom faces.*

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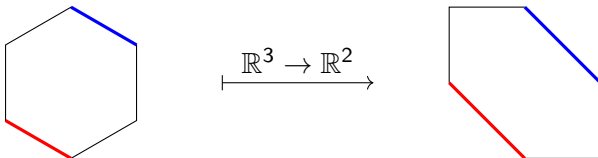
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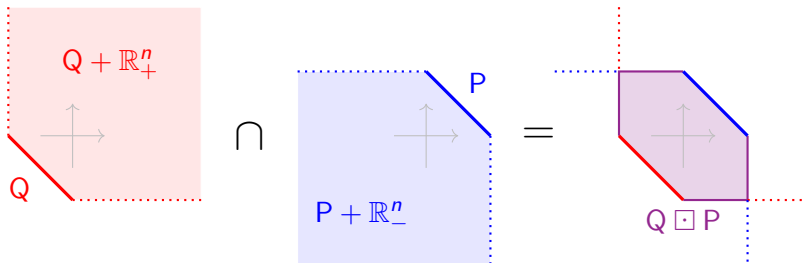
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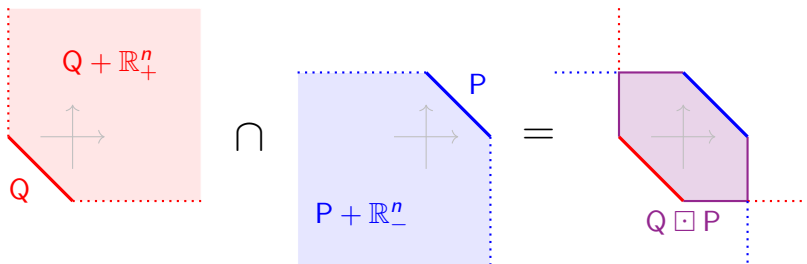
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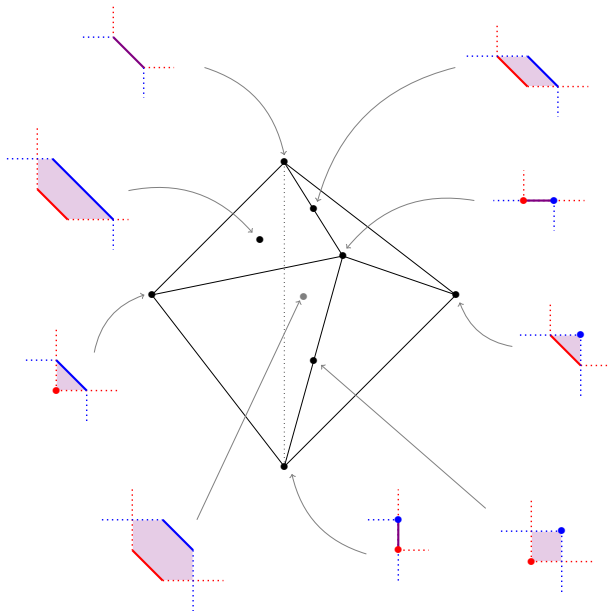
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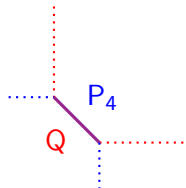
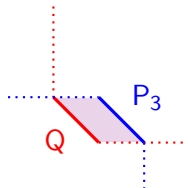
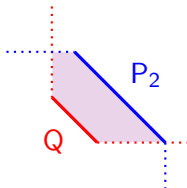
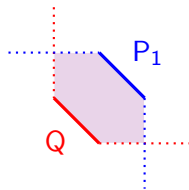
Reciprocal of $P \mapsto (P^+, P^-)$ is the map $(P, Q) \mapsto Q \boxdot P$

SC_3 drawn with GP-sums



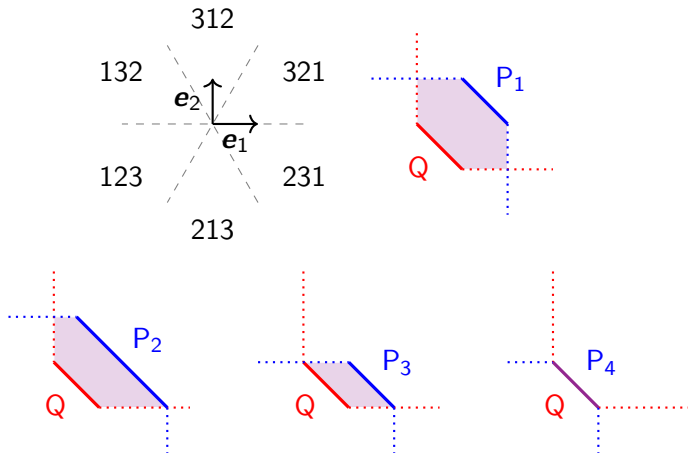
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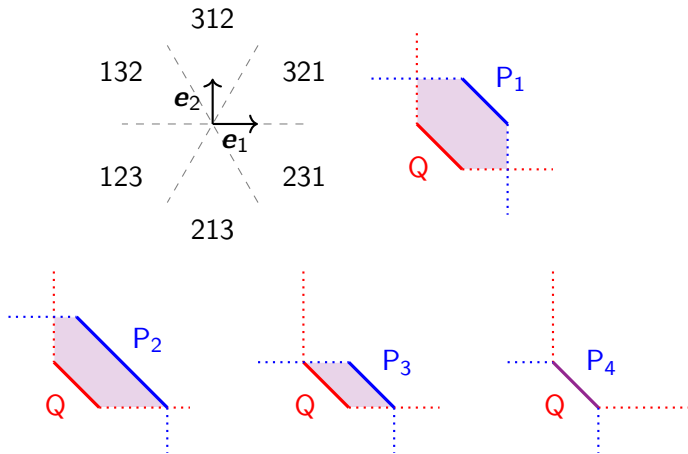
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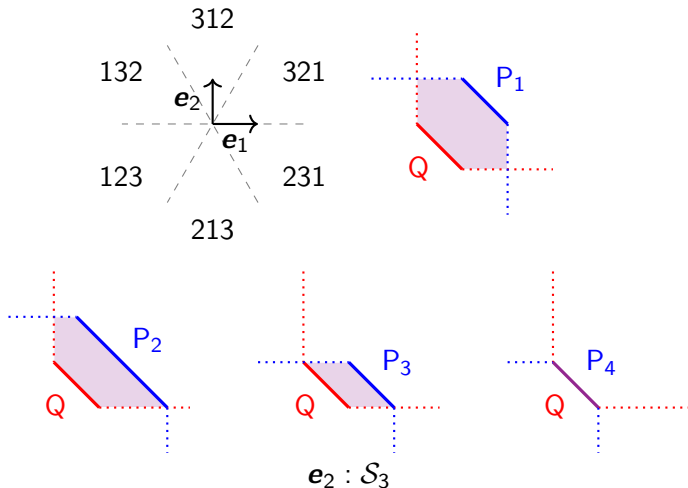
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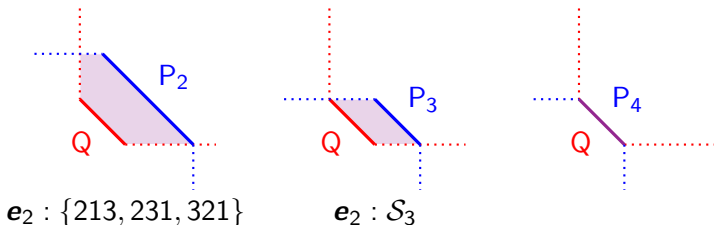
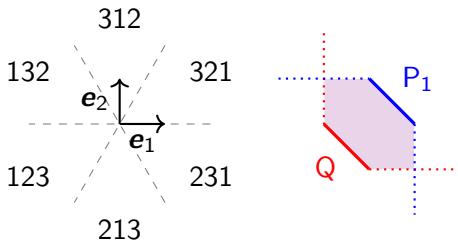
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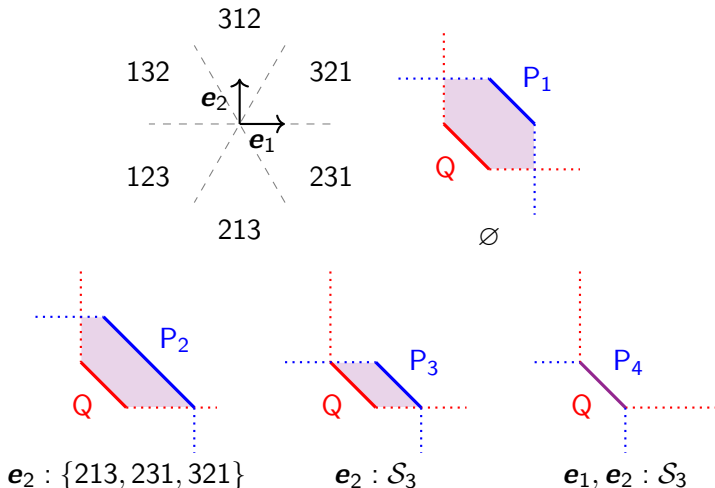
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Poset of valid equality sets

$$P, Q \in \mathbb{SC}_n$$

Definition

Equality set: $\mathbf{E}(P, Q) := \{(i, \sigma) \in [n] \times \mathcal{S}_n ; Q_i^\sigma = P_i^\sigma\}$

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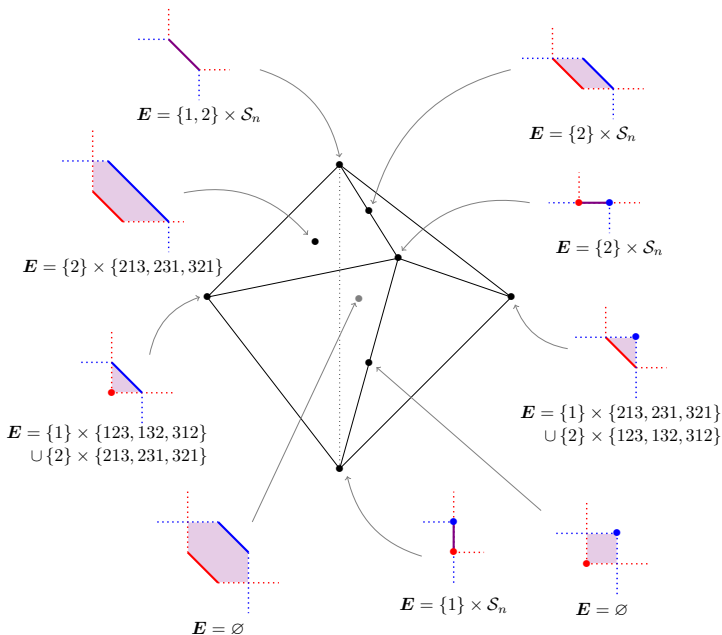
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SC_3 with equality sets



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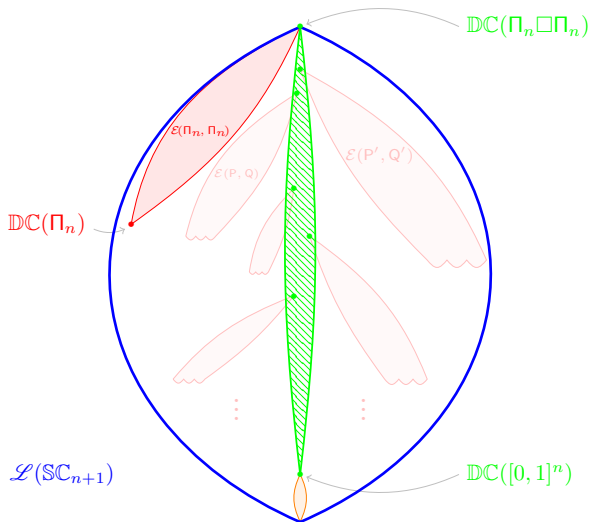
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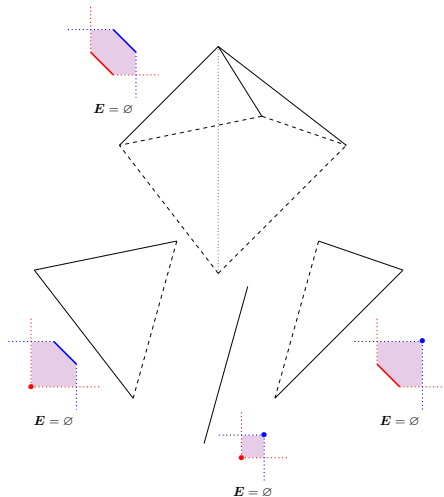
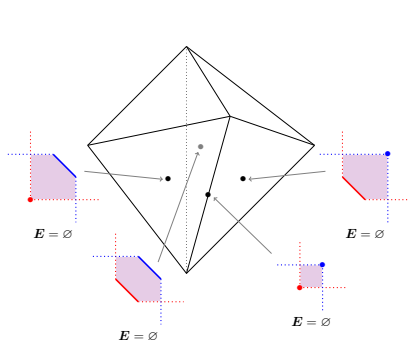
Corollary (Loho–Padrol–P., 25+)

The faces of $\mathbb{SC}_{n+1}$ are in bijection with triples $(P, Q, \mathbf{E})$ where P and Q are taken up to equivalence and $\mathbf{E} \in \mathcal{E}(P, Q)$.

$\mathbb{DC}(P)$: class of $P' \sim P$, i.e. faces of $\mathbb{SC}_n$

$$\mathcal{L}(\mathbb{SC}_n)^2 \simeq \{\mathbb{DC}(Q \square P) ; P, Q \in \mathbb{SC}_n^2\}$$

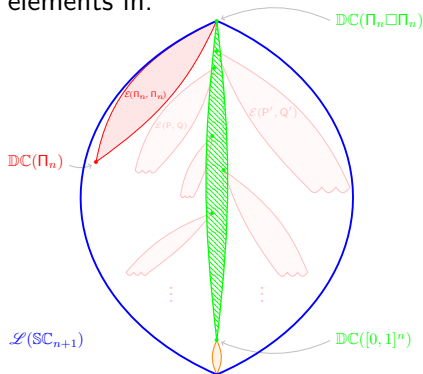




Applications (now, we can count stuff)

Rays

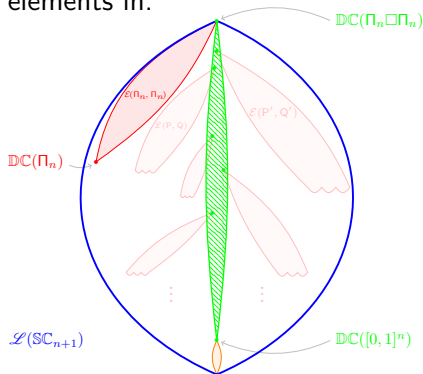
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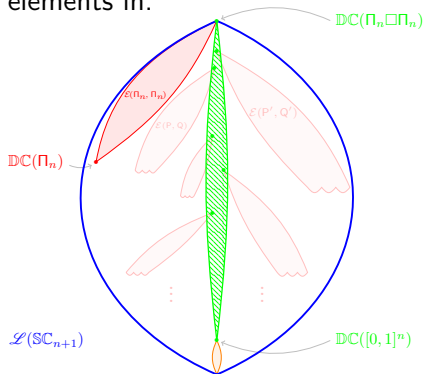


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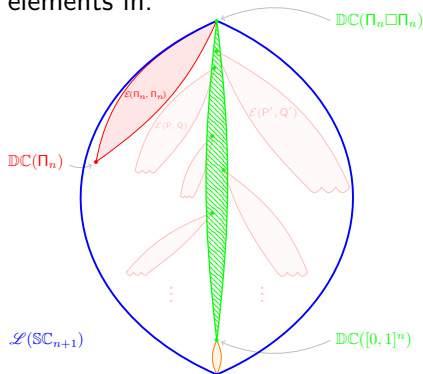


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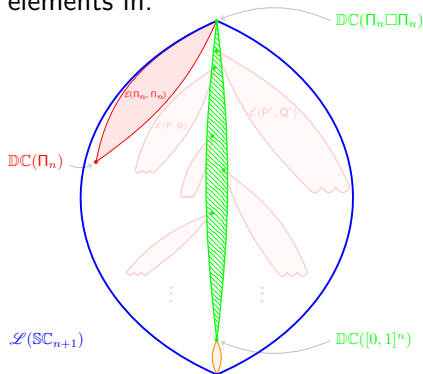


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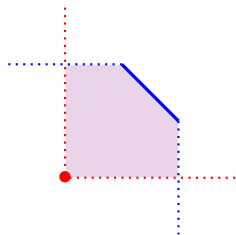
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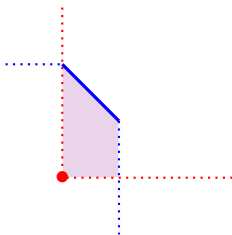
To increase hopes, start with $P, Q \in \text{rays of } \mathbb{SC}_n$!

Rays

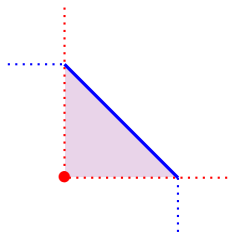
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$E = \emptyset$



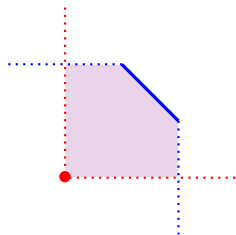
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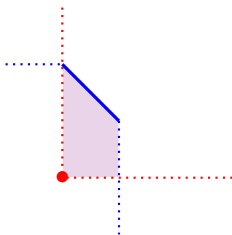
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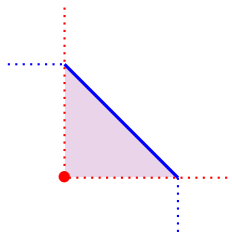
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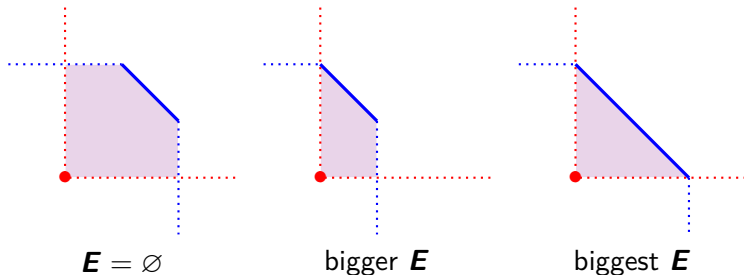


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Problem: Sometimes, you'll need to dilate by 0 to get a ray.

Lemma (Loho–Padrol–P., 25+)

Technical lemma about non-0 dilations that you wouldn't read.

Theorem (Loho–Padrol–P., 25+)

If $P_1, \dots, P_s$ are pair-wise fertile rays of $\mathbb{SC}_n$, then you can construct explicitly $R_1, \dots, R_{s^2}$ pair-wise fertile rays of $\mathbb{SC}_{n+1}$.

Number of rays of $\mathbb{SC}_n$

t_n := number of rays of $\mathbb{SC}_n$.

Theorem ((approximately) Nguyen, '86)

The number t_n of rays of $\mathbb{SC}_n$ satisfies

$$2^{2^n/n^{3/2}} \leq t_n$$

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Theorem (Loho–Padrol–P., '25+)

The number t_n of rays of $\mathbb{SC}_n$ satisfies

$$2^{2^{n-2}} \leq t_n$$

N.B.: By Upper Bound Theorem: $t_n \leq n^{2^n}$.

Precisely: $n - 2 \leq \log_2 \log_2 t_n \leq n + \log_2 \log_2 n + 1$

Lemma (Loho–Padrol–P., 25+)

$$\dim \mathbb{DC}(Q \square P) = \dim \mathbb{DC}(P) + \dim \mathbb{DC}(Q) + 1$$

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$F_n(X) := \sum_k f(n, k) X^k$ where $f(n, k) :=$ number k -faces of $\mathbb{SC}_n$

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$$\text{Hence, } F_{n+1}(X) \succeq_{\text{coeff.-wise}} F_4^{2^{n-4}}$$

$$\text{And, } \#_{\text{faces}} \mathbb{SC}_n \geq 22\,108^{2^{n-4}}$$

Lemma (Loho–Padrol–P., 25+)

$\mathbb{DC}(P)$ *not simplicial* $\Rightarrow \mathbb{DC}(Q \square P)$ *not simplicial* for any $Q \in \mathbb{SC}_n$.

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Corollary (Loho–Padrol–P., 25+)

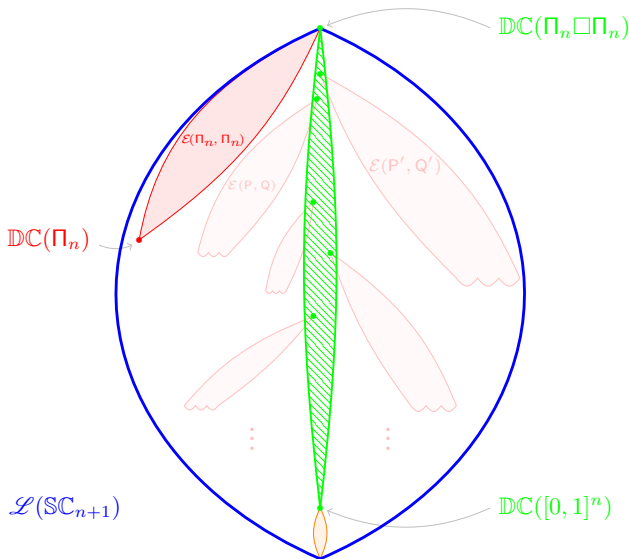
Number of non-simplicial faces of $\mathbb{SC}_n \geq \prod_{p < n} \#_{\text{faces}} \mathbb{SC}_p$

In particular, $\mathbb{SC}_n$ is “very far” from being simplicial, so we should be “very far” for the Upper Bound Theorem equality case.

Furthermore:

- $\mathcal{L}(\mathbb{SC}_n)^2 \simeq \{\mathbb{DC}(Q \square P); P, Q \in \mathbb{SC}_n^2\}$ is the face figure of $[0, 1]^n$.
- $\mathcal{E}(P, P)$ is iso. to the interval of $\mathcal{L}(\mathbb{SC}_{n+1})$ between P and $P \square P$. In particular, $\mathcal{E}(\Pi_n, \Pi_n)$ is a face figure.
- In general, $\mathcal{E}(P, Q)$ is a co-graded, co-atomistic, join-semi-lattice with a geometric interpretation.
- Notion of independence polytope Ind_P for all generalized permutohedra, similar to matroid polytopes, with:
 Ind_P indecomposable iff P indecomposable, and
 $\text{Ind}_{P+Q} = \text{Ind}_P + \text{Ind}_Q$

Thank you!

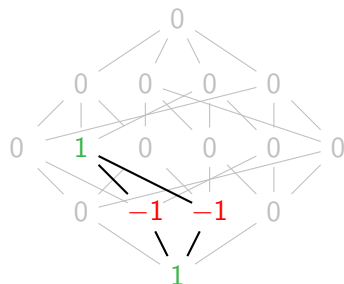
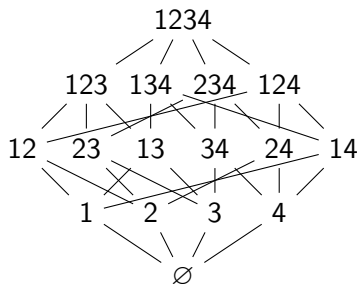


Tools: submodular dependencies

Notations: $Sx = S \cup \{x\}$, $(\mathbf{f}_x)_{x \subseteq [n]}$ canonical basis of $\mathbb{R}^{2^{[n]}}$

Definition

Submodular vector $\mathbf{n}(S, u, v) = \mathbf{f}_{Suv} - \mathbf{f}_{Su} - \mathbf{f}_{Sv} + \mathbf{f}_S$
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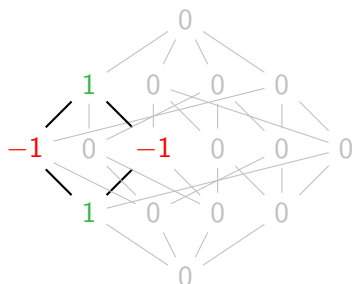
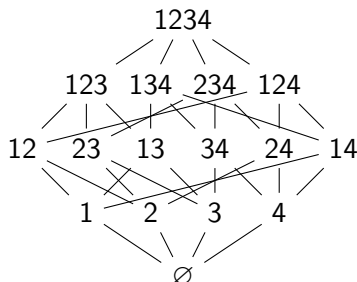


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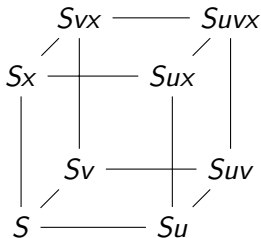
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NB: Cubic relations generates all relations of submodular vectors

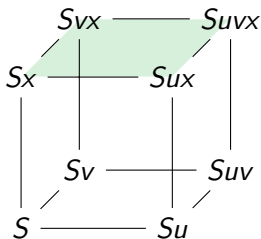
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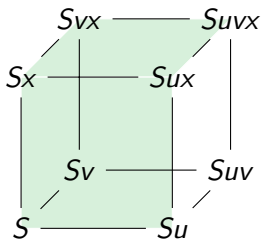
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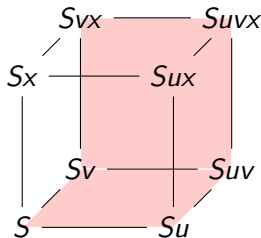
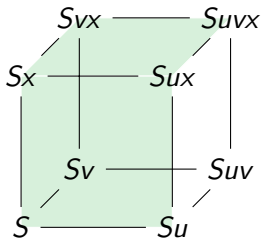
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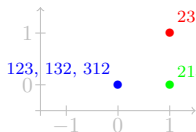
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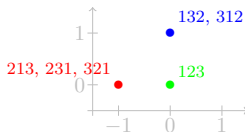
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Fertility

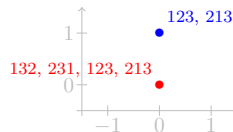
(P, Q) is *fertile* iff there exists $i \in [n]$ such that for all $\tau \in S_n$, either $P_i^\tau \neq \min_{\sigma \in S_n} P_i^\sigma$ or $Q_i^\tau \neq \max_{\sigma \in S_n} Q_i^\sigma$



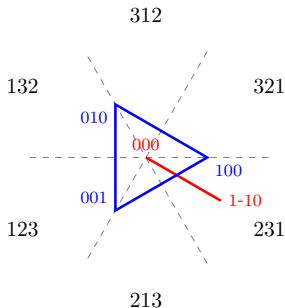
$i = 1$



$i = 2$



$i = 3$



Multiple choice questions (sometimes several answer possible)

A sum of 2 co-planar triangles can have	3 edges	4 edges	5 edges	6 edges
$\mathcal{N}_{P+Q} = \text{--- of } \mathcal{N}_P, \mathcal{N}_Q$	union	intersection	common refinement	product
Edge directions of $GP \in$	$\{\mathbf{e}_i\}_i$	$\{\mathbf{e}_i + \mathbf{e}_j\}_{i,j}$	$\{\mathbf{e}_i - \mathbf{e}_j\}_{i,j}$	$\{\mathbf{e}_i \pm \mathbf{e}_j\}_{i,j}$
Z_G has no hexagonal face iff G is	a tree	K_3 -free	K_4 -free	complete
--- form a basis of $\text{Vect}(\mathbb{SC}_n)$	$\{\Delta_X\}_{X \subseteq [n]}$	nestohedra	shard polytopes	matroid polytopes
$\dim \mathbb{SC}_3 =$	3	4	5	11
Group of symmetries of $\mathbb{SC}_n$:	Free_n	$\mathcal{S}_n$	$\mathcal{S}_n \times \mathbb{Z}_2$	$\mathcal{S}_n \times \mathbb{Z}_n$
# 3-dim indecomposable GP:	3	5	7	37

1. Show that $\text{Vert}(P + Q) \subseteq \{\mathbf{u} + \mathbf{v} ; \mathbf{u} \in \text{Vert}(P), \mathbf{v} \in \text{Vert}(Q)\}$. Find an example with strict inclusion. Prove that $\mathcal{N}_{P+Q}$ is the common refinement of $\mathcal{N}_P$ and $\mathcal{N}_Q$.
2. Find all the ways to write the 2-dimensional permutahedron as a Minkowski sum of indecomposable polytopes.
3. Prove that $\mathbf{h}_{P+Q} = \mathbf{h}_P + \mathbf{h}_Q$ and $\ell_{P+Q} = \ell_P + \ell_Q$. What are $\mathbf{h}_{P+t}$ and ℓ_{P+t} ?
4. Prove that a triangle is Minkowski indecomposable. Deduce that all simplicial polytopes are Minkowski indecomposable.

5. Take vectors $\mathbf{s}, \mathbf{s}', \mathbf{r}_1, \dots, \mathbf{r}_{d-1}$ such that $C = \text{cone}(\mathbf{s}, \mathbf{r}_1, \dots, \mathbf{r}_{d-1})$ and $C' = \text{cone}(\mathbf{s}', \mathbf{r}_1, \dots, \mathbf{r}_{d-1})$ are simplicial cones which intersect on their proper face $C \cap C' = \text{cone}(\mathbf{r}_1, \dots, \mathbf{r}_{d-1})$. Show that there exist $\alpha_{\mathbf{s}}, \alpha_{\mathbf{s}'} > 0$ and $\alpha_{\mathbf{r}_i} \in \mathbb{R}$ such that:

$$\alpha_{\mathbf{s}}\mathbf{s} + \alpha_{\mathbf{s}'}\mathbf{s}' + \sum_i \alpha_{\mathbf{r}_i}\mathbf{r}_i = \mathbf{0}$$

6. Compute (and draw) the deformation cone of a parallelogram (use the edge-lengths point of view). What about a parallelepiped? Show that there is a **unique** way to write a parallelepiped P as a sum of Minkowski indecomposable polytopes.

7. Show that $\mathbb{SC}_n$ has $\binom{n}{2}2^{n-2}$ facets. Give bounds on its number of rays.

8. Show that graphical zonotopes are generalized permutahedra. Show that nestohedra are generalized permutahedra. Show that matroid polytopes are generalized permutahedra. Which are indecomposable?

9. If G is triangle-free, then $\mathbb{DC}(Z_G)$ is simplicial (see lecture). Each face of $\mathbb{DC}(Z_G)$ is associated to a polytope: which polytope?

10. The *weighted graphical zonotope* of a graph $G = (V, E)$ and weight (on edges) $\omega : E \rightarrow \mathbb{R}_+^*$ is $Z_{G,\omega} := \sum_{(i,j) \in E} \omega(i,j)[\mathbf{e}_i, \mathbf{e}_j]$. Show that: $\{\text{zonotopes}\} \cap \{\text{generalized permutahedra}\} = \{\text{weighted graphical zonotopes}\}$. Show that the cone of weighted graphical zonotopes is a section (i.e. intersection with a linear sub-space) of the submodular cone, such that the rays of this section are rays of $\mathbb{SC}_n$. What is the dimension of this section?

11. Prove the cubic relations hold. Show that cubic relations generates all linear relations between submodular vectors.